Data Science
Complete All Unit Semester Exam Notes

Unit	Coverage
Unit 1	Introduction, lifecycle, applications, security, preprocessing and EDA
Unit 2	Python libraries, supervised/unsupervised learning, regression, classification and optimization
Unit 3	Deep learning, frameworks, CNN, style transfer, text generation and time series
Unit 4	R, RStudio, data structures, visualization, regression and classification
Unit 5	Analytics tools, version control and case studies

UNIT 1: Introduction to Data Science, Preprocessing and EDA

1. Meaning, Scope and Evolution of Data Science

Data Science is an interdisciplinary field that uses scientific methods, statistical techniques, programming, machine learning and domain knowledge to extract meaningful insights from data. It converts raw data into useful information, predictions and decisions. In modern organizations, Data Science is used not only for reporting but also for automation, forecasting and intelligent decision making.

The evolution of Data Science can be understood as a gradual development from statistics to modern AI-based analytics. Initially, data was analyzed using statistical tools and spreadsheets. Later, database systems and data warehouses helped store large datasets. Data mining introduced automatic pattern discovery. Big data technologies enabled processing of huge and fast data. Today, Data Science combines all these with machine learning, deep learning and cloud computing.

Data Science has become important because every field generates data. Business transactions, sensors, mobile apps, social media, healthcare systems, banking systems and educational portals produce large amounts of data. This data has value only when it is cleaned, analyzed and converted into insight.

- Statistics gives mathematical foundation for analysis.
- Programming helps clean, process and automate data workflows.
- Machine learning helps build predictive models.
- Domain knowledge helps interpret results correctly.
- Visualization helps communicate insights to decision makers.

Period/Stage	Main Idea	Contribution to Data Science
Statistics	Mathematical analysis of data	Mean, variance, probability, hypothesis testing
Database Era	Structured storage and querying	SQL, data warehouses and reports
Data Mining	Pattern discovery	Classification, clustering, association rules
Big Data	Processing large and fast data	Hadoop, Spark, distributed storage
Modern Data Science	End-to-end insight and prediction	ML, AI, visualization, deployment

2. Data Science Roles

A Data Science project is usually completed by a team because it requires different types of expertise. Some people focus on business understanding, some on data engineering, some on analysis and some on model deployment. In small teams, one person may perform multiple roles.

The Data Analyst focuses on reports, dashboards and business insights. The Data Scientist builds statistical and machine learning models. The Data Engineer creates pipelines and manages storage. The Machine Learning Engineer converts models into production systems. The Business Analyst understands business requirements and communicates results to stakeholders.

- Data Analyst: prepares reports, dashboards and descriptive analysis.
- Data Scientist: performs EDA, feature engineering, modeling and interpretation.
- Data Engineer: builds ETL pipelines, databases and data lakes.
- ML Engineer: deploys and monitors machine learning models.
- Domain Expert: validates whether the result is practically meaningful.

Role	Responsibility	Tools/Skills
Data Analyst	Reporting, dashboarding, descriptive statistics	Excel, SQL, Power BI, Python/R
Data Scientist	Model building and insight discovery	Python, R, ML, statistics
Data Engineer	Data pipelines and storage	SQL, Spark, ETL, cloud
ML Engineer	Model deployment and optimization	APIs, Docker, MLOps
Business Analyst	Requirement and decision support	Domain knowledge, communication

A Data Science project follows a lifecycle. The first stage is problem definition, where the objective, target users, expected output and success metric are decided. Without a clear problem statement, the model may solve the wrong problem.

The second stage is data collection. Data can be collected from databases, files, APIs, surveys, sensors or web sources. After collection, data preprocessing is performed to clean missing values, duplicates, noisy values and inconsistent formats. Exploratory Data Analysis then helps understand trends, correlations and outliers.

After EDA, features are selected or engineered and models are trained. The model is evaluated using suitable metrics. If performance is acceptable, it is deployed into a dashboard, API or application. After deployment, monitoring is required because data patterns can change over time.

- Problem definition decides what needs to be solved.

- Modeling creates predictive or descriptive solutions.

3. Stages in a Data Science Project

- Data collection gathers relevant data.
- Preprocessing improves data quality.
- EDA identifies patterns and relationships.

- Evaluation checks model quality.
- Deployment makes the solution usable.
- Monitoring ensures performance over time.

legal problems.

4. Applications and Data Security Issues

Data Science is used in almost every field. In healthcare, it can predict disease risk and analyze medical images. In banking, it detects fraud and supports credit scoring. In e-commerce, it recommends products and predicts demand. In education, it predicts student performance and identifies students who need support.

Data security is important because Data Science projects often use sensitive data such as names, phone numbers, health records, financial transactions and user behavior. If data is leaked or misused, it can cause privacy loss, financial loss and

- Use encryption for data at rest and in transit.
- Use access control and authentication.
- Use anonymization or masking for personal data.
- Do not collect unnecessary data.
- Maintain audit logs and follow ethical data practices.

Field	Application	Example
Healthcare	Prediction and diagnosis	Diabetes prediction

Finance	Risk and fraud analysis	Credit card fraud detection
Retail	Recommendation and forecasting	Product recommendation
Education	Student analytics	Pass/fail prediction
Field	**Application**	**Example**
Agriculture	Yield and weather analysis	Crop yield forecasting

5. Data Preprocessing Techniques

Data preprocessing is the process of converting raw data into clean and useful data. It is one of the most important stages because real-world data is incomplete, noisy and inconsistent. Poor quality data leads to poor quality models, even if advanced algorithms are used.

Data cleaning handles missing values, duplicate records, outliers and wrong entries. Data integration combines data from multiple sources. Data transformation converts data into suitable format using scaling, normalization or encoding. Data reduction decreases data size while preserving meaning. Data discretization converts continuous values into intervals.

- Missing values can be removed or filled using mean, median or mode.
- Outliers can be detected using box plots or z-score.
- Categorical variables can be converted using label encoding or one-hot encoding.

Technique	Purpose	Example
Cleaning	Improve quality by removing errors	Remove duplicate student records
Integration	Merge multiple datasets	Join attendance and marks
Transformation	Change format or scale	Normalize salary values
Reduction	Reduce size/features	PCA, sampling
Discretization	Convert numeric to bins	Age group: young/adult/senior

6. Exploratory Data Analysis and Descriptive Statistics

Exploratory Data Analysis is the process of studying data before applying models. It helps understand distributions, missing values, outliers, correlations and patterns. EDA uses both numerical summaries and visual tools.

Descriptive statistics summarize data. Mean gives the average value. Standard deviation measures spread around the mean. Skewness shows whether data is symmetric or tilted toward one side. Kurtosis shows peakness and tail behavior. Box plots show median, quartiles and outliers. Pivot tables summarize data by categories. Heat maps show values or correlations using colors.

- It detects data quality problems.
- It supports feature engineering.
- Scaling makes numerical variables comparable.
- Feature selection removes irrelevant variables.

- EDA reduces wrong assumptions about data.

- It helps identify important variables.

- It helps communicate data patterns visually.

Term	Formula/Meaning	Interpretation
Mean	Sum of values / number of values	Central average
Standard Deviation	Spread around mean	High SD means high variation
Skewness	Asymmetry of distribution	Positive or negative tilt

Term	Formula/Meaning	Interpretation
Kurtosis	Tail heaviness/peakness	High kurtosis means heavy tails
Box Plot	Median, quartiles, outliers	Detects spread and outliers
Heat Map	Color-coded matrix	Shows correlations/intensity

UNIT 2: Python for Data Science

1. Python and Important Libraries

Python is widely used in Data Science because it is simple, readable and has a powerful ecosystem of libraries. It supports data collection, cleaning, analysis, visualization, machine learning and deep learning. Python also works well with Jupyter Notebook and Google Colab, which are useful for interactive experimentation.

NumPy provides fast numerical arrays and mathematical operations. Pandas provides Series and DataFrame for tabular data.
Scikit-learn provides machine learning algorithms, preprocessing tools and evaluation metrics. Visualization libraries such as

- NumPy is used for arrays, matrices and linear algebra.
- Pandas is used for data cleaning and tabular manipulation.
- Scikit-learn is used for regression, classification and clustering.
- Matplotlib and Seaborn are used for visualization.
- Python supports complete data science workflow.

Library	Main Object/Feature	Use
NumPy	ndarray	Numerical computing and matrix operations
Pandas	Series, DataFrame	Data cleaning, filtering, grouping
Scikit-learn	Estimator API	ML models and evaluation
Matplotlib	Figure, Axes	Basic charts and plots
Seaborn	Statistical plots	Heat maps, pair plots, box plots

2. Supervised Learning: Regression and Classification

Supervised learning uses labeled data. The algorithm learns from input-output pairs and predicts output for new data. If the output is continuous, the problem is regression. If the output is a category, the problem is classification.

In regression, examples include predicting house price, sales or marks. In classification, examples include spam detection, disease prediction and pass/fail prediction. The model is trained on training data and tested on unseen test data.

Aspect	Regression	Classification
Output	Continuous value	Class label
Example	Predict salary	Spam or not spam
Algorithms	Linear Regression, Random Forest Regressor	Logistic Regression, SVM, Decision Tree
Metrics	MAE, MSE, RMSE, R2	Accuracy, Precision, Recall, F1

Matplotlib and Seaborn help create charts and graphs.

- Regression output is numerical and continuous.
- Classification output is a class label.
- Training data teaches the model.
- Testing data checks generalization.
- Metrics must match the problem type.

3. Ordinary Least Squares Regression

Ordinary Least Squares is a linear regression technique. It tries to find the best-fitting line by minimizing the sum of squared differences between actual and predicted values. These differences are called residuals.

OLS is easy to understand and interpret. The coefficient of a feature shows how much the target changes when that feature increases by one unit, assuming other features are constant. However, OLS is sensitive to outliers and may perform poorly for non-linear data.

- Equation: y = b0 + b1x1 + b2x2 + error.
- Goal: minimize squared residuals.

- Assumes independent errors and low multicollinearity.

4. Logistic Regression, SVM and Ensemble Methods

Logistic Regression is used for classification. It predicts probability using the sigmoid function. If probability is greater than threshold, the sample is assigned to a class. It is useful for binary classification and is easy to interpret.

Support Vector Machine finds the best decision boundary that separates classes with maximum margin. Ensemble methods combine multiple models to improve performance. Random Forest uses bagging, while boosting methods build models sequentially to correct previous errors.

- SVM works well in high-dimensional feature spaces.
- Random Forest reduces overfitting by averaging many trees.

- Ensembles are powerful but may be less interpretable.

Algorithm	Type	Main Idea
Logistic Regression	Classification	Uses sigmoid to predict class probability
SVM	Classification/Regression	Finds maximum margin boundary
Decision Tree	Both	Splits data using feature rules
Random Forest	Ensemble	Combines many trees using bagging
Gradient Boosting	Ensemble	Sequentially improves weak learners

5. Unsupervised Learning and Evolutionary Optimization

Unsupervised learning works with unlabeled data. It is used to discover hidden patterns and structures. Clustering groups similar data points. Dimensionality reduction reduces the number of features while preserving useful information.

- Works best when relationship is linear.

- Useful for simple prediction and interpretation.

- Logistic Regression outputs probability.

- Boosting improves weak models step by step.

Evolutionary optimization is inspired by natural evolution. It starts with a population of possible solutions. Each solution is evaluated using a fitness function. Better solutions are selected, combined and mutated to create new solutions. This is useful when the search space is large.

- K-Means is a common clustering algorithm.
- Hierarchical clustering creates tree-like grouping.
- PCA is used for dimensionality reduction.
- Evolutionary techniques use selection, crossover and mutation.
- They are useful for complex optimization problems.

UNIT 3: Deep Learning and Time Series Analysis

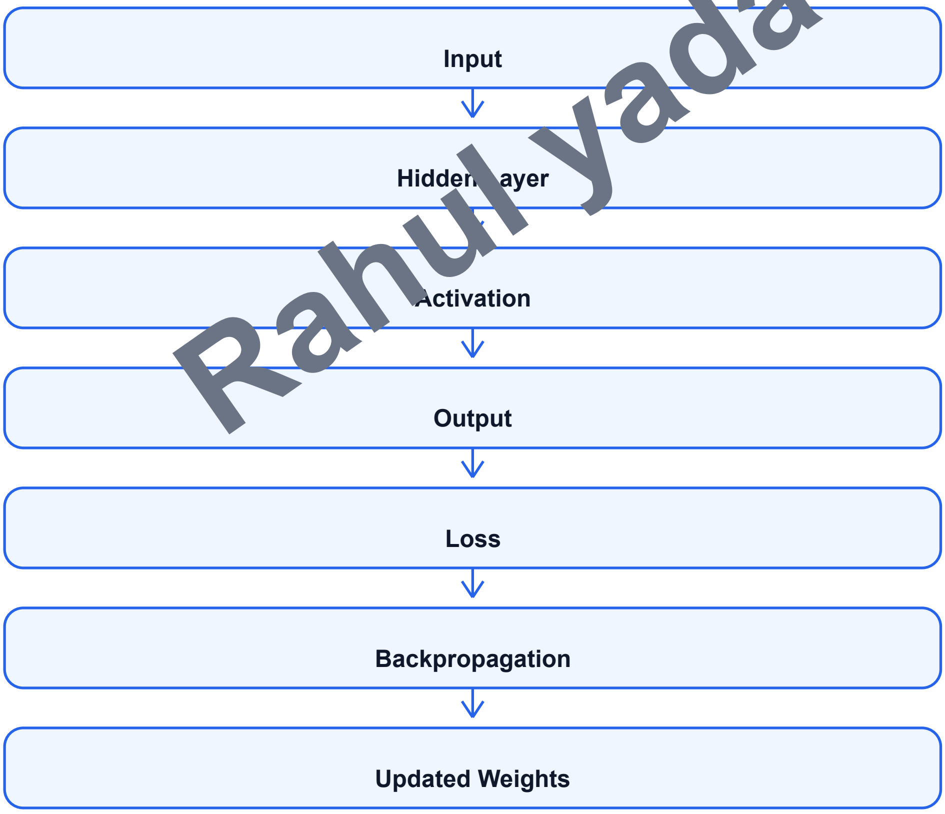
1. Deep Learning Fundamentals

Deep Learning is a branch of machine learning based on artificial neural networks with multiple layers. These layers learn representations from data automatically. It is useful for complex data such as images, text, speech and time series.

A neural network contains neurons, weights, bias, activation functions, loss function and optimizer. During training, the model makes predictions, calculates error and updates weights using backpropagation. Deep networks can learn complex non-linear patterns.

Deep learning usually needs large datasets and high computational power. GPUs are often used to speed up training. Although deep learning gives high accuracy, it may be difficult to interpret compared to simpler models.

- Input layer receives data.
- Hidden layers learn intermediate features.
- Output layer produces prediction.
- Activation functions introduce non-linearity.
- Loss function measures prediction error.
- Optimizer updates weights to reduce loss.



2. TensorFlow, Keras and PyTorch

TensorFlow is an open-source framework for machine learning and deep learning. It is suitable for large-scale training and production deployment. Keras is a high-level API that simplifies neural network building using layers, models, optimizers and callbacks.

PyTorch is another popular framework. It is known for dynamic computation graphs and easy debugging. Researchers often prefer PyTorch because it is flexible and Python-friendly. Both TensorFlow and PyTorch support GPU acceleration.

- TensorFlow is strong for production and deployment.
- Keras is simple and beginner-friendly.
- PyTorch is flexible and research-friendly.

- All support CNN, RNN, transformers and custom models.prediction.

- Choice depends on project needs and team skill.

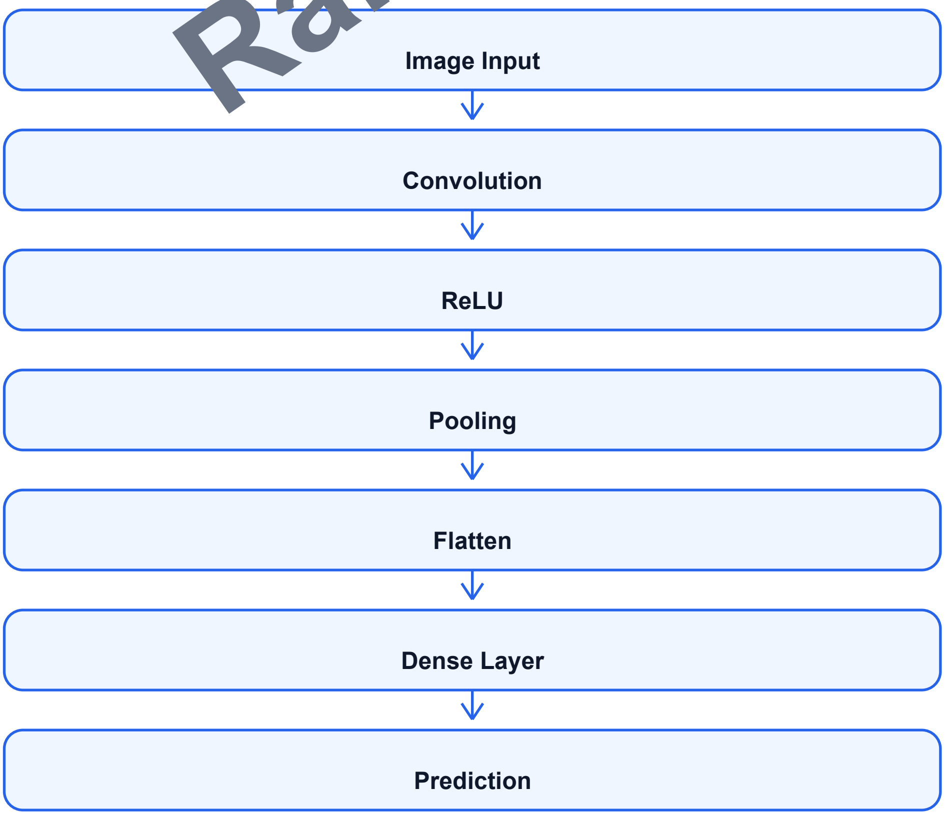
Point	TensorFlow/Keras	PyTorch
Graph	Static/optimized execution	Dynamic graph
Ease	Keras is very simple	Pythonic and flexible
Deployment	Strong tools like TF Serving/TFLite	TorchScript and serving tools
Use	Production and standard workflows	Research and custom models

3. CNN, Style Transfer, Text Generation and Sentiment Analysis

Convolutional Neural Networks are mainly used for image-related tasks. They use filters to detect visual patterns such as edges, corners and textures. Pooling layers reduce the size of feature maps and fully connected layers perform final

Style transfer uses deep learning to combine content of one image with the style of another image. Text generation creates new text by predicting the next word or character. Sentiment analysis classifies opinions as positive, negative or neutral.

- CNN is useful in image classification and object detection.
- Style transfer combines content representation and style representation.
- Text generation is used in chatbots and writing assistants.
- Sentiment analysis is used in review and social media analysis.
- PyTorch can implement image and text deep learning models.



4. Time Series Analysis

Time series data is collected in time order. Examples include daily temperature, monthly sales, stock prices and electricity demand. Time series analysis studies patterns over time and forecasts future values.

Important components are trend, seasonality, cyclic variation and noise. Trend is long-term increase or decrease. Seasonality is repeated pattern at fixed intervals. Noise is random variation. Before modeling, time series is plotted and checked for stationarity.

- Trend shows long-term movement.

- Stationarity means mean and variance remain stable.

5. ARMA, ARIMA, Fourier Analysis and State Space Models

ARMA combines autoregressive and moving average components. Autoregressive part uses past values, while moving average part uses past errors. ARMA works for stationary time series. ARIMA extends ARMA by adding differencing, which helps make non-stationary data stationary.

Fourier analysis decomposes a signal into frequency components. It is useful when periodic behavior exists. Spectral estimation studies how signal power is distributed over frequency. State-space models represent hidden states that evolve over time and produce observed data.

- ARIMA uses differencing for non-stationary series.

- State-space models are useful for hidden dynamic systems.

Model/Concept	Meaning	Use
AR	Uses previous values	Short-term dependence
MA	Uses previous errors	Noise correction
ARMA	AR + MA	Stationary time series
ARIMA	ARMA + differencing	Non-stationary forecasting
Fourier	Frequency decomposition	Seasonal/periodic signals
State Space	Hidden state representation	Dynamic systems

- Seasonality repeats after fixed time period.

- Noise is random fluctuation.

- Forecasting predicts future values.

- AR model uses past values.
- MA model uses past errors.

- ACF and PACF help identify model order.
- Fourier analysis is useful for periodic signals.

UNIT 4: R for Data Science

1. R and RStudio

R is a programming language and environment designed for statistical computing, data analysis and visualization. It is widely used in academics, research, statistics and analytics. R has many packages for data manipulation, visualization and machine learning.

RStudio is an integrated development environment for R. It provides console, script editor, plots panel, files panel, package

- RStudio improves productivity and project management.

- R is useful for reports, dashboards and statistical models.

Data structures are used to store data in different forms. R provides vectors, factors, lists, arrays, matrices and data frames. Understanding these structures is important because every analysis task requires data to be stored correctly.

A vector stores one-dimensional same-type data. A factor stores categorical data. A list can store different types of elements. A matrix stores two-dimensional same-type data. A data frame stores table-like data where each column can have different type. Data frames are most commonly used in data science.

Structure	Dimension	Data Type	Example
Vector	1D	Same	c(10,20,30)
Factor	1D	Categories	Low, Medium, High
List	Mixed	Different	list(name, marks, model)
Array	Multi-D	Same	3D numeric data
Matrix	2D	Same	Numerical table
Data Frame	2D	Column-wise different	CSV dataset

manager and environment viewer. This makes coding, debugging and visualization easier.

- R is strong in statistics and visualization.

- R packages extend functionality.

2. R Data Structures

- Vector: one-dimensional and same data type.
- Factor: categorical variable.
- List: collection of different objects.
- Array: multidimensional same-type data.
- Matrix: two-dimensional same-type data.
- Data frame: tabular data with columns.

3. Working with Data and Visualization in R

Working with data in R includes importing, inspecting, cleaning, transforming and visualizing data. Data can be imported from CSV, Excel, databases and web sources. After importing, functions like head(), str(), summary() and dim() help understand the dataset.

Visualization helps identify distributions, relationships and outliers. R supports base plotting and advanced visualization using ggplot2. Histograms show distribution, scatter plots show relationship, box plots show spread and outliers, and line charts show trends over time.

- read.csv() imports CSV data.
- head() displays first records.
- str() shows data structure.

- summary() gives descriptive statistics.classification.

- ggplot2 creates professional plots.

Chart	Purpose	Example Use
Histogram	Distribution of numerical data	Marks distribution
Bar Chart	Compare categories	Students per branch
Scatter Plot	Relationship between variables	Hours studied vs marks
Box Plot	Median, quartiles, outliers	Salary spread
Line Plot	Trend over time	Monthly sales

4. Regression and Classification with R

R provides built-in and package-based tools for regression and classification. Linear regression can be performed using lm(), which models a continuous target variable. Logistic regression can be performed using glm() with binomial family for binary

For advanced classification, packages like caret, randomForest and e1071 are used. A typical modeling workflow includes importing data, splitting train/test data, training model, predicting and evaluating results.

- caret provides common interface for model training.

- Regression predicts continuous values.
- Classification predicts categories.
- lm() is used for linear regression.
- glm() is used for logistic regression.

Point	Python	R
Strength	General-purpose ML and deployment	Statistics and visualization
Data Structure	Pandas DataFrame	R Data Frame
Visualization	Matplotlib, Seaborn	ggplot2
Modeling	Scikit-learn	caret, lm, glm
Users	Developers and ML engineers	Statisticians and analysts

UNIT 5: Data Analytics Software and Case Studies

1. Overview of Data Analytics Software

Data analytics software helps users process, analyze, visualize and model data. Some tools are code-based, while others provide graphical user interfaces. Such tools are useful because they reduce manual work and make analytics accessible to more users.

Weka, Orange and RapidMiner are popular for machine learning and data mining. Minitab is widely used for statistics and

- Some tools support no-code or low-code workflows.

- Tool choice depends on task, skill and organization needs.

Tool	Purpose	Important Features
Weka	Data mining and ML	GUI, classification, clustering, preprocessing
Orange	Visual analytics	Drag-and-drop workflows, widgets
RapidMiner	End-to-end analytics	Data prep, ML, deployment
Minitab	Statistical analysis	Quality control, hypothesis testing
Power BI	Business intelligence	Dashboards, reports, DAX
GitHub	Version control	Repositories, branches, collaboration
Google Colab	Cloud notebooks	Python, GPU/TPU, sharing

2. Version Control for Data Science Projects

Version control tracks changes in files over time. In Data Science, code, notebooks, reports and configuration files change frequently. Version control helps reproduce results and collaborate safely.

Git is the most common version control system, and GitHub is a popular online platform for hosting repositories. Data Science projects should store scripts, notebooks, README files and model configuration in Git. Large sensitive datasets should not be

- Supports team collaboration through branches.

- Improves reproducibility of experiments.

quality control. Power BI is used for business intelligence dashboards. GitHub supports version control and collaboration. Google Colab provides cloud-based notebooks for Python and machine learning.

- Analytics tools simplify data preparation.
- They support visualization and reporting.

- They help share results with stakeholders.directly committed.

- Tracks history of code and notebooks.
- Allows rollback to older versions.

- Secrets and private data must not be committed.

3. Case Study: Student Performance Prediction

This project predicts student performance using academic and behavioral data. Input variables may include attendance, internal marks, assignment submission, quiz scores, previous semester marks and participation. The target may be final marks, pass/fail or risk category.

The project begins with data collection from college systems. Data is cleaned to handle missing marks and inconsistent enrollment numbers. EDA checks relation between attendance and marks. A classification model like logistic regression or random forest can predict whether a student is at risk. The result can be shown in a dashboard for teachers.

- Problem type: classification or regression.
- Data: attendance, marks, assignments, quizzes.
- Models: logistic regression, decision tree, random forest.
- Output: risk level or predicted marks.
- Benefit: early support for weak students.

4. Case Study: Sales Forecasting

Sales forecasting predicts future demand using historical sales data. The data may contain date, product, store, price, discount, festival season and advertising information. Since the data is time-based, time series methods are commonly used.

EDA is performed to identify trend and seasonality. ARIMA, moving average or regression models can be used. Forecasting helps businesses manage inventory, plan marketing and reduce stock problems. Power BI dashboards can show actual sales, predicted sales and category-wise trends.

- Problem type: time series forecasting.
- Data: date-wise sales, product, price, discount.
- Models: ARIMA, regression, moving average.
- Visualization: line chart and forecast plot.
- Benefit: better inventory and business planning.

5. Case Study: Sentiment Analysis

Sentiment analysis identifies opinion from text data. It can be applied to product reviews, social media posts, feedback forms and support tickets. The output is usually positive, negative or neutral sentiment.

The project starts with text cleaning, tokenization and stopword removal. Text is converted into numerical features using TF-IDF or embeddings. A classification model such as Naive Bayes, Logistic Regression or deep learning model is trained. Results help companies understand customer satisfaction.

- Problem type: text classification.
- Data: reviews, comments, tweets or feedback.
- Preprocessing: cleaning, tokenization, stopword removal.
- Models: Naive Bayes, Logistic Regression, LSTM.
- Benefit: customer opinion and product improvement.

Quick Revision: Definitions and Differences

Term	Exam Definition
Data Science	Field that extracts insights and predictions from data using statistics, programming and domain knowledge.
EDA	Process of exploring data using statistics and visualizations before modeling.
Data Cleaning	Removing or correcting errors, missing values, duplicates and noisy data.
Regression	Supervised learning technique to predict continuous values.
Classification	Supervised learning technique to predict class labels.
Clustering	Unsupervised learning technique to group similar data points.
Deep Learning	Machine learning based on multilayer neural networks.
Time Series	Data collected in time order.
Power BI	Business intelligence tool for dashboards and reports.
Version Control	Tracking and managing changes in project files.

Important Comparisons

Comparison	Point 1	Point 2
Data Mining vs Data Science	Data mining finds patterns	Data Science includes full lifecycle and prediction
R vs Python	R is strong in statistics	Python is strong in ML and deployment
ARMA vs ARIMA	ARMA needs stationary data	ARIMA handles non-stationary data using differencing
BoW-like sparse features vs Embeddings	Sparse features are simple	Embeddings capture meaning better
Dashboard vs Report	Dashboard is interactive	Report is usually static

Important Semester Exam Questions

Unit 1

- Explain evolution of Data Science in detail.
- Explain roles in a Data Science project.
- Explain stages of Data Science lifecycle with diagram.
- Explain data preprocessing techniques.

- Explain descriptive statistics and EDA tools.

Unit 2

- Explain Python libraries used in Data Science.

- Differentiate regression and classification.

- Explain Ordinary Least Squares Regression.

Unit 3

- Compare TensorFlow, Keras and PyTorch.

- Explain CNN with diagram.

Unit 4

- Explain R and RStudio.

- Explain R data structures with examples.

- Explain importing and visualizing data in R.

- Explain regression and

- Explain Logistic Regression, SVM and Ensemble Methods.

- Explain unsupervised learning and evolutionary optimization.

- Explain deep learning and neural network architecture.

- Explain time series components and forecasting.

- Explain ARMA, ARIMA, Fourier analysis and state-space models.

- Explain Weka, Orange, RapidMiner and Minitab.

- Explain version control in Data Science projects.

classification in R.- Compare Python and R for Data Science.

Unit 5

- Explain Power BI, GitHub and Google Colab.

- Write a case study on student performance prediction.

- Write a case study on sales forecasting or sentiment analysis.

How to Write Long Answers in Exam

- Start with a clear definition.

- Draw a diagram or flowchart whenever possible.

- Write 4-6 explained points.

- Add a table/comparison for theory questions.

- End with example or application.